

RIM — Rollback & Integrity Module

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Category: Governance & Enforcement

Module: Rollback & Integrity Module (RIM)

Type: Recovery & Integrity Control Module

Standard: Self-Evolving Systems Control Standard (SESCS)

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Status: Canonical Module

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Authority: OOF™ Origin Open Foundation™

Protection: MIP™ — Methodological Intellectual Property

Canonical Language: English (UCL™)

Compatibility:

SESCS · MCM · MLM · MVM · RAM · TSA · MTVF · EVIP · ArtData · AI Governance Systems · ORGS

Canonical Definition System

Rollback & Integrity Module (RIM) defines the mandatory capability of a system to restore previous valid states and maintain structural integrity after mutations or system failures.

Base

Without rollback and integrity control:

- system errors become permanent
- damage cannot be reversed
- system degradation accumulates
- recovery is impossible

A system that cannot recover cannot be trusted.

Protocol

Any system **MUST**:

- maintain recoverable system states
- enable rollback to previous valid state
- verify integrity after rollback
- prevent propagation of corrupted states

Absence of rollback results in:

Irreversible System State (ISS)

Scope

RIM applies to:

- AI systems with evolving logic
 - autonomous decision systems
 - adaptive infrastructures
 - multi-agent systems
 - runtime mutation environments
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Methodology

1. capture system state
 2. store valid checkpoints
 3. detect failure or anomaly
 4. initiate rollback
 5. restore previous state
 6. verify system integrity
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Rollback Conditions

Rollback MUST be triggered when:

- mutation validation fails post-deployment
 - system behavior becomes unstable
 - contamination is detected (via TSA)
 - system integrity is compromised
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Integrity Verification

After rollback, system MUST:

- confirm structural consistency
 - verify functional stability
 - validate alignment with system rules
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Structural Rule (Critical)

No system mutation is complete without rollback capability.

Checkpoint Requirement

System MUST maintain:

- defined recovery points
 - state history
 - integrity checkpoints
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MIF — Minimal Implementation Framework

Step 1 — State Capture

Capture system state.

Minimum:

- defined checkpoint

Step 2 — Storage

Store recoverable state.

Minimum:

- persistent state record

Step 3 — Rollback Mechanism

Enable restoration.

Minimum:

- revert capability

Step 4 — Integrity Check

Verify restored system.

Minimum:

- basic consistency check

MIF+ — Extended Implementation Framework

Includes all MIF requirements plus:

- automated rollback triggers
 - multi-state recovery layers
 - distributed checkpoint storage
 - real-time integrity monitoring
 - integration with TSA contamination detection
 - predictive failure rollback
 - continuous system integrity validation
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Decision Logic

System state is:

RECOVERABLE → rollback possible

IRREVERSIBLE → rollback not possible

Use Case 1 — AI System Failure Recovery

Scenario

AI system mutation introduces instability.

Problem

- system behaves unpredictably
- no recovery mechanism

Application

RIM enables:

- rollback to stable state
- integrity verification

Result

- system restored
 - failure contained
 - risk minimized
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Use Case 2 — Contaminated System State

Scenario

System becomes contaminated (TSA classification).

Problem

- corrupted behavior spreads
- system cannot self-correct

Application

RIM ensures:

- rollback to clean state
- prevention of contamination propagation

Result

- system integrity restored
 - contamination removed
 - safe operation resumed
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Compatibility

RIM operates with:

- MCM — defines mutation boundaries
 - MLM — provides mutation history
 - MVM — validates mutations
 - RAM — assigns responsibility
 - TSA — detects contamination
 - MTVF — validation framework
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Structural Principle

A system must be able to return to a valid state.

Closing Statement

Change without recovery creates irreversible risk.
Irreversible risk destroys system integrity.

Rollback & Integrity Module (RIM) establishes a non-negotiable condition:

Every system must be able to recover, restore, and verify its integrity.

Related Documents

[→ SESCO Standard](#)

[→ SESCO — Master Index](#)

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Canonical Source

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